



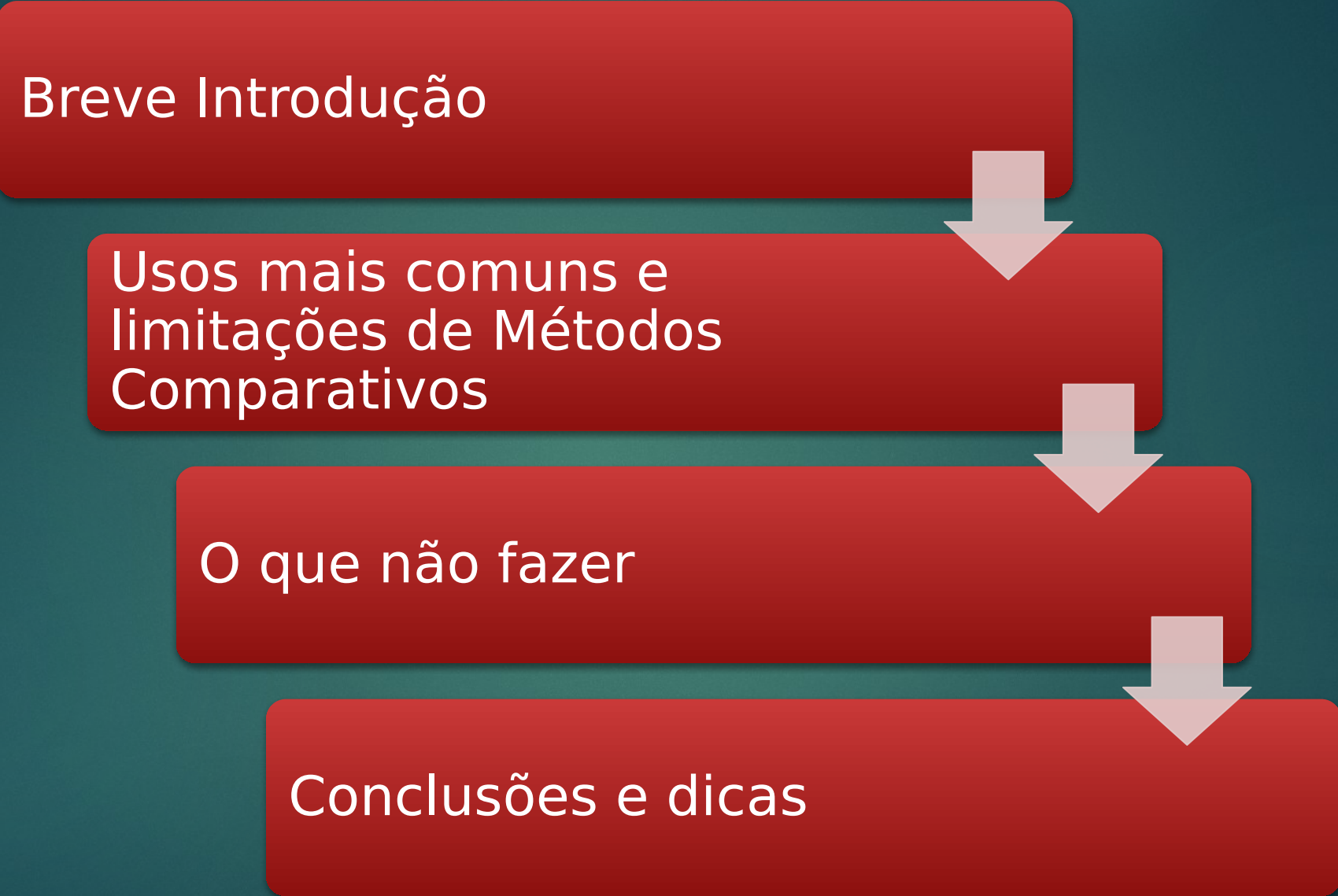
Usos e Desusos de Métodos Filogenéticos Comparativos

DR. DIOGO B. PROVETE

dbprovete@gmail.com

diogoprovete.weebly.com

Breve Introdução



```
graph TD; A[Breve Introdução] --> B[Usos mais comuns e limitações de Métodos Comparativos]; B --> C[O que não fazer]; C --> D[Conclusões e dicas];
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Usos mais comuns e
limitações de Métodos
Comparativos

O que não fazer

Conclusões e dicas

O que são Métodos Filogenéticos Comparativos?

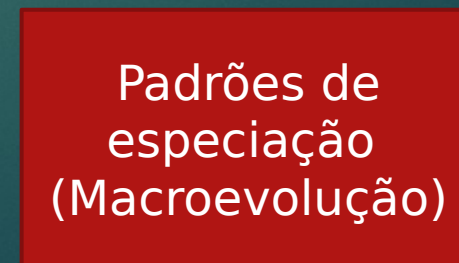
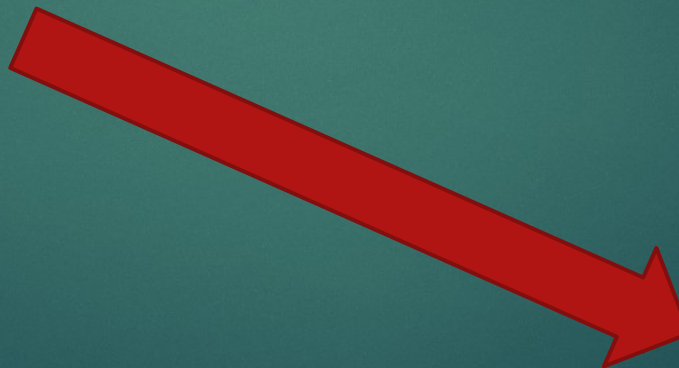
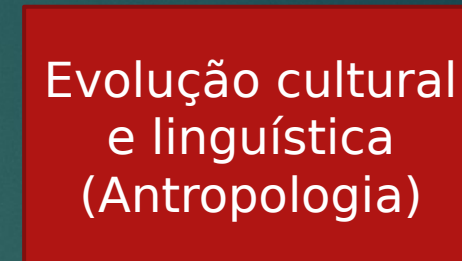
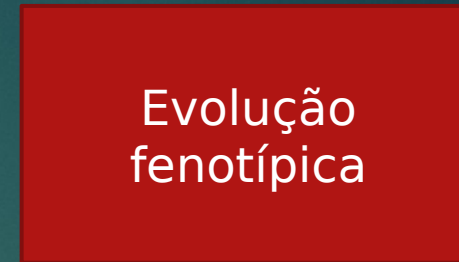
- ▶ Conjunto de métodos estatísticos que usam a relação de parentesco entre as espécies para testar hipóteses evolutivas sobre seus atributos
- ▶ Interesse surgiu no fim da década de 1970 com a disponibilidade de filogenias
 - ▶ Processo evolutivo deixa "rastros" no fenótipo das espécies
 - ▶ Espécies mais próximas *tendem* a ser mais parecidas
- ▶ Perguntas comuns
 - ▶ Qual o padrão filogenético de uma dada característica das espécies?
 - ▶ Sob que processo e em que velocidade essa característica evolui?
 - ▶ Qual a correlação dessa característica com outras ou com a variação ambiental?

```
graph LR; A([Métodos Filogenéticos Comparativos]) --> B[Evolução fenotípica]; A --> C[Padrões de especiação (Macroevolução)];
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Métodos
Filogenéticos
Comparativos

Evolução
fenotípica

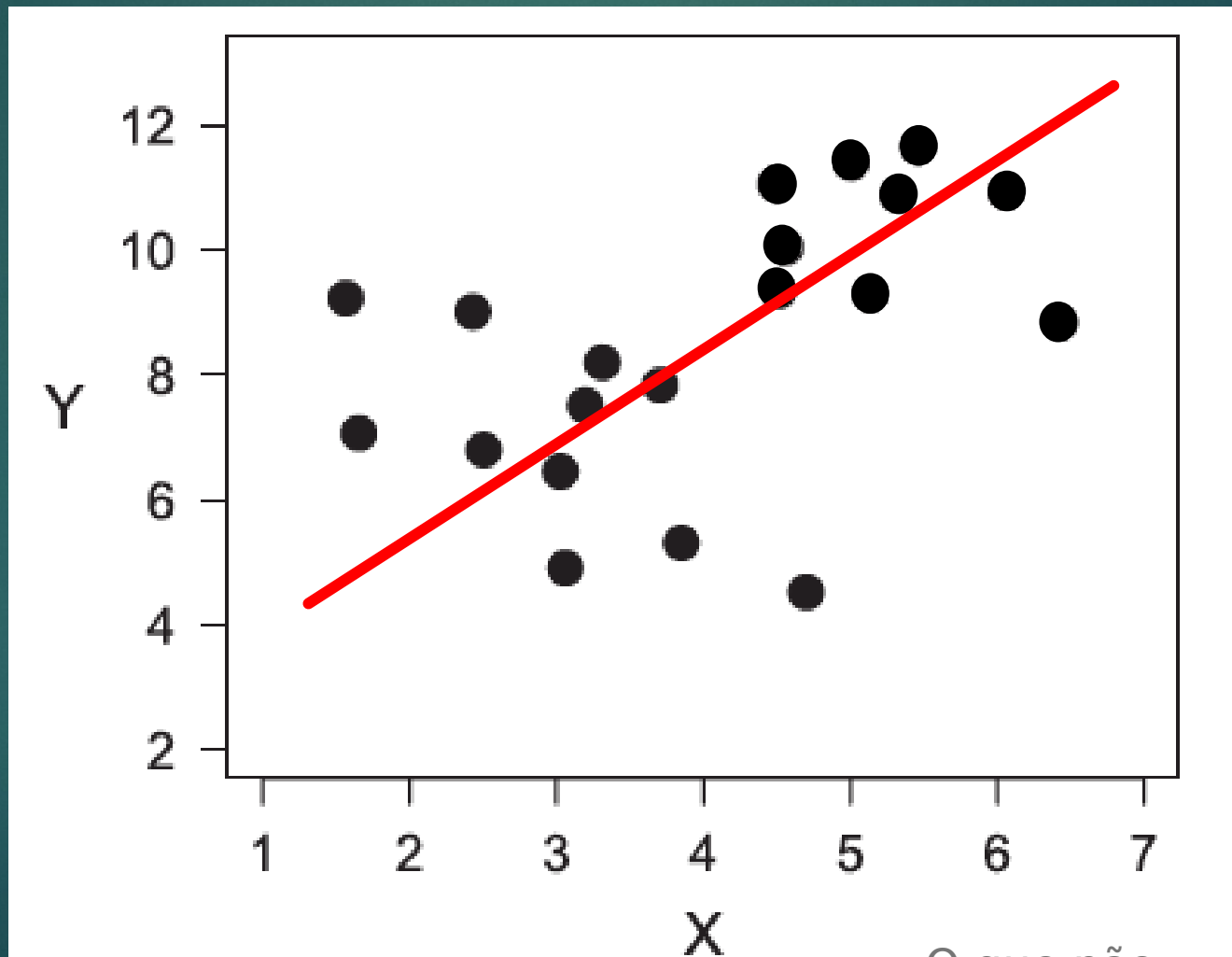
Padrões de
especiação
(Macroevolução)



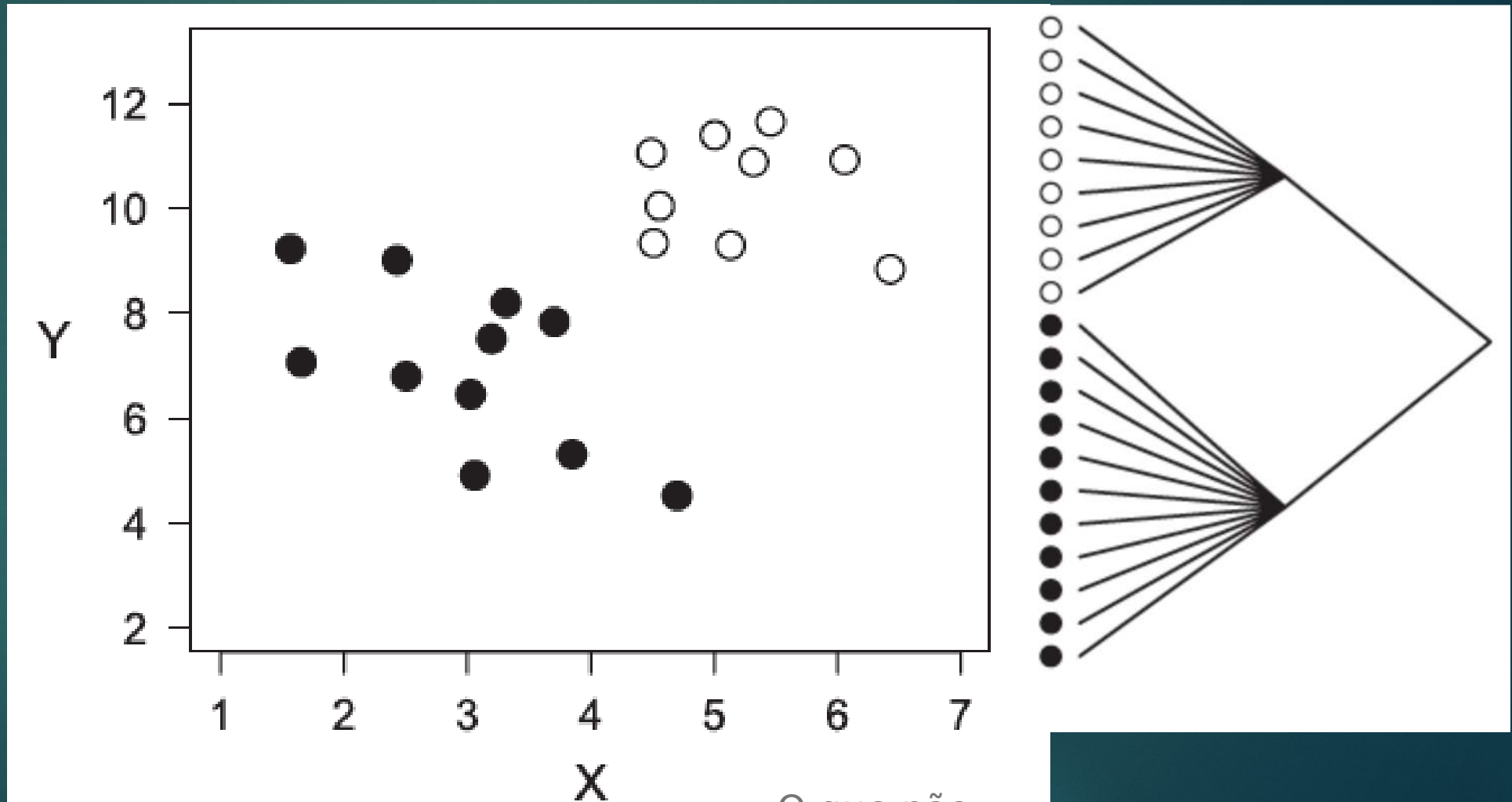
O que são Métodos Filogenéticos Comparativos?

- ▶ Problema da Autocorrelação Filogenética => viola os pressupostos dos métodos "convencionais"
- ▶ Necessidade de se incorporar a relação entre as espécies nas análises

O problema do parentesco comum



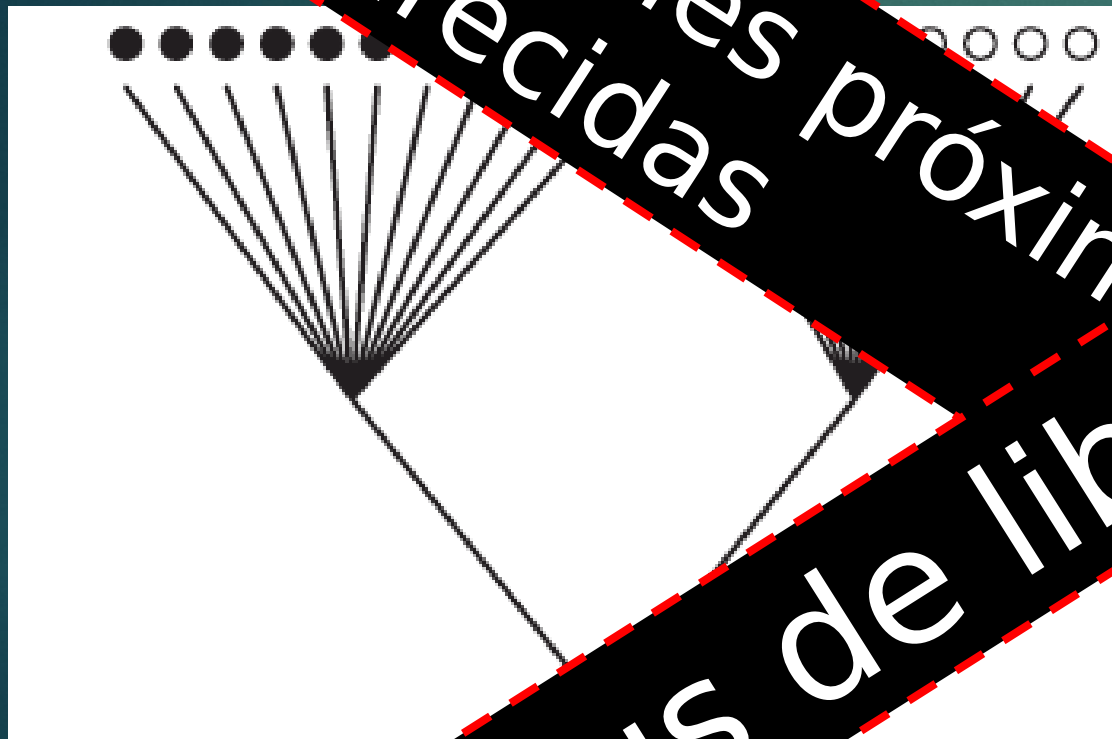
O problema do parentesco comum



Espécies não são independentes

Espécies próximas parecidas

Graus de liberdade são mais



em



“Phylogenies are
fundamental to
evolutionary biology, there
is no doing it without taking
them into account”

Joe Felsenstein (1985)



Evolution, 39(6), 1985, pp. 1335–1351

THE QUANTITATIVE ASSESSMENT OF PHYLOGENETIC CONSTRAINTS IN COMPARATIVE ANALYSES: SEXUAL DIMORPHISM IN BODY WEIGHT AMONG PRIMATES

JAMES M. CHEVERUD

*Departments of Anthropology, Cell Biology & Anatomy, and Ecology & Evolutionary Biology,
Northwestern University, Evanston, IL 60201*

MALCOLM M. DOW

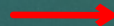
*Department of Anthropology and Program in Mathematical Methods in the Social Sciences,
Northwestern University, Evanston, IL 60201*

AND

WALTER LEUTENEGGER

Department of Anthropology, University of Wisconsin, Madison, WI 53706

1985



Vol. 125, No. 1

The American Naturalist

January 1985

PHYLOGENIES AND THE COMPARATIVE METHOD

JOSEPH FELSENSTEIN

Department of Genetics SK-50, University of Washington, Seattle, Washington 98195

Submitted November 30, 1983; Accepted May 23, 1984

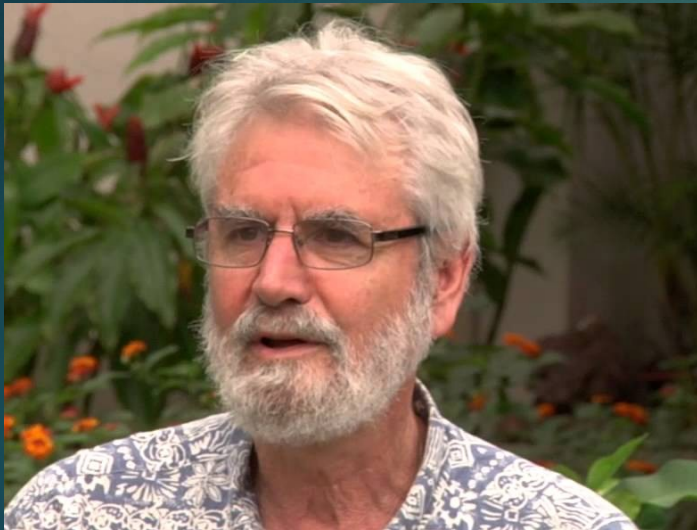


VOL. 193, NO. 6 THE AMERICAN NATURALIST JUNE 2019

HISTORICAL COMMENT

Revisiting a Key Innovation in Evolutionary Biology: Felsenstein's "Phylogenies and the Comparative Method"

Raymond B. Huey,^{1,*} Theodore Garland Jr.,² and Michael Turelli³



1987

Evolution, 41(5), 1987, pp. 1098–1115

PHYLOGENETIC STUDIES OF COADAPTATION: PREFERRED
TEMPERATURES VERSUS OPTIMAL PERFORMANCE
TEMPERATURES OF LIZARDS

RAYMOND B. HUEY

Department of Zoology NJ-15, University of Washington, Seattle, WA 98195
AND

ALBERT F. BENNETT

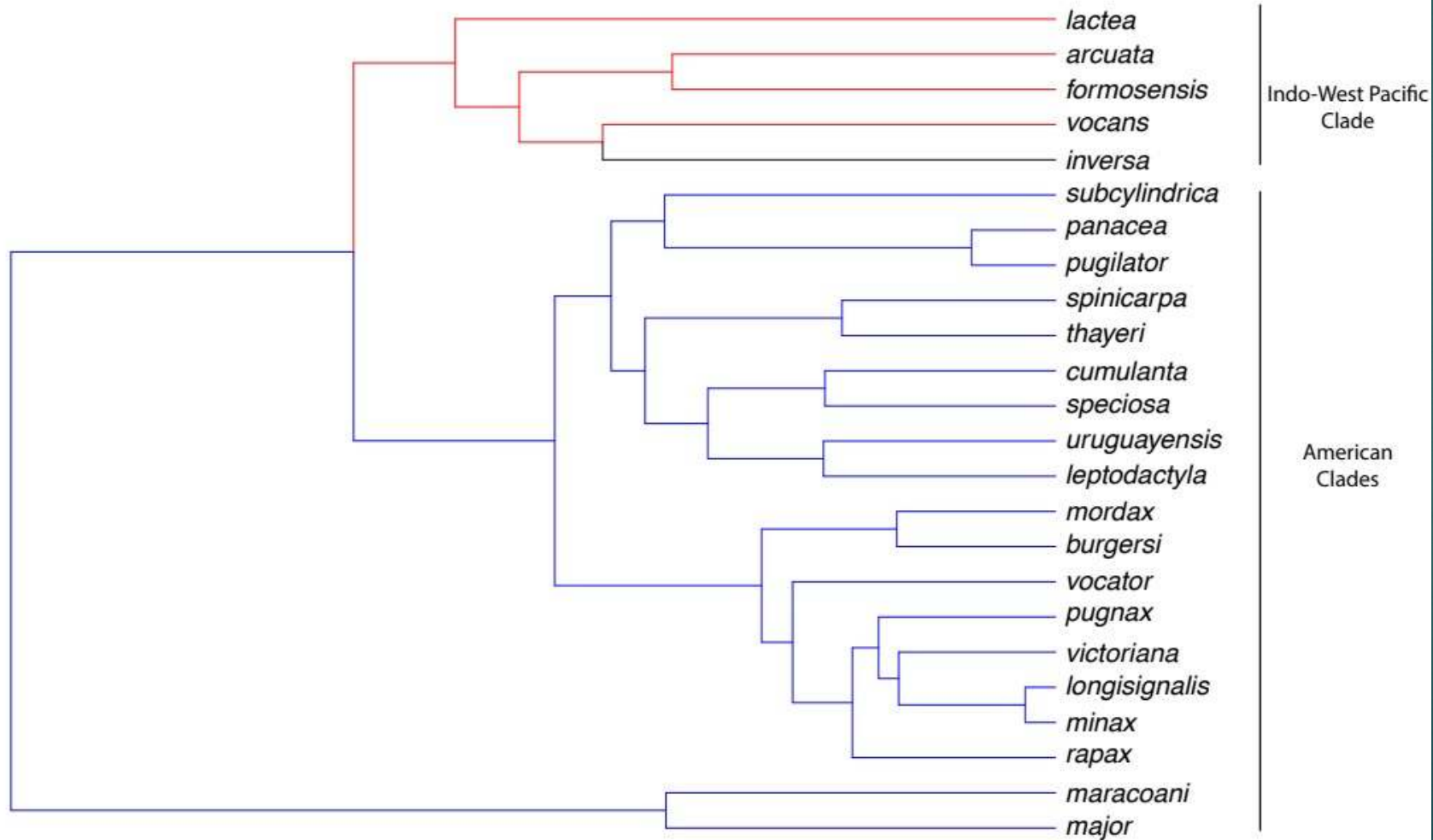
School of Biological Sciences, University of California, Irvine, CA 92717

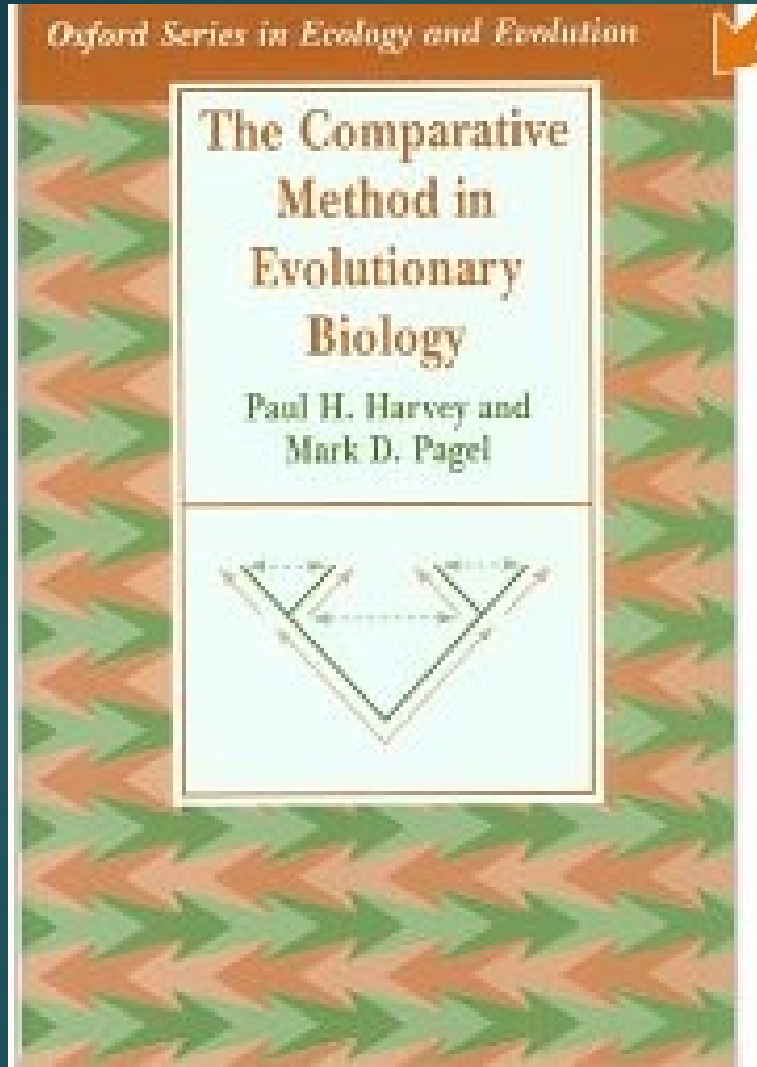
Estimativa de estado ancestral: Squared-change parsimony

Principais métodos/Abordagens

- ▶ Estimativa ("reconstrução") de estado ancestral
- ▶ Mensuração e teste de "sinal" filogenético
- ▶ Ajuste de Modelos de Evolução à atributos (BM, OU, EB/LB, trend etc)
- ▶ Estimativa de taxas de evolução de atributo(s) ao longo da filogenia
- ▶ Estimativa de ótimos/picos/regimes adaptativos ao longo da filogenia
- ▶ Evolução correlacionada de atributos contínuos (PGLS, phyloGLM) e categóricos (Pagel's method)

- 1 Does salinity drive physiological evolution? Phylogenetic patterns and the adaptive
- 2 landscape of osmoregulation in fiddler crabs (Brachyura, *Uca*)
- 3
- 4 Samuel Coelho Faria^{1*}, Diogo Borges Provete², Carl Leo Thurman³, and John
- 5 Campbell McNamara^{1,4}

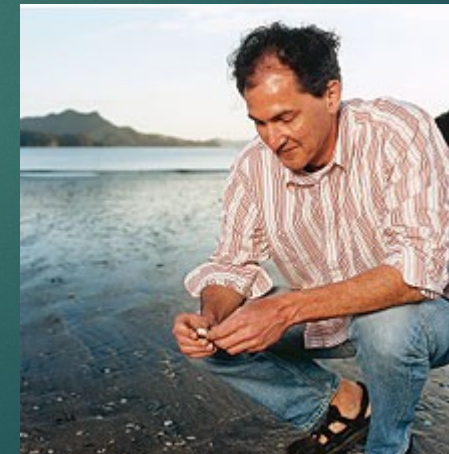




1991

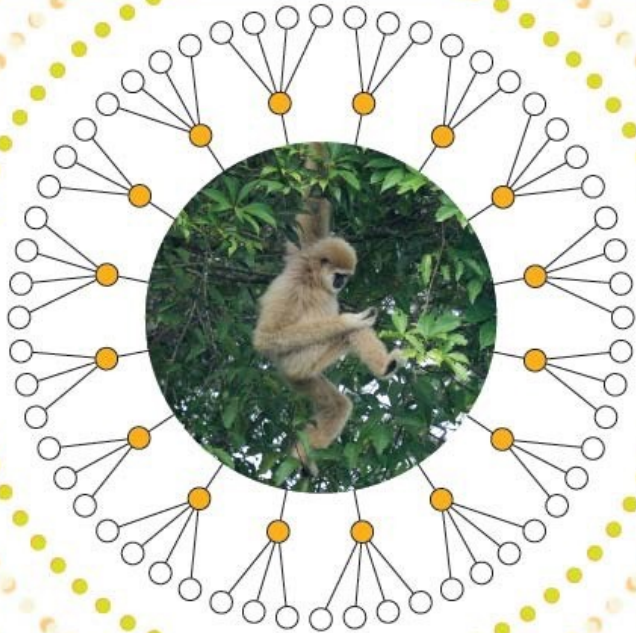


Paul Harvey



Mark Pagel

The **Comparative Approach**
in **Evolutionary Anthropology**
and **Biology**



Charles L. Nunn



Charles Nunn, Duke University

2011

Apresentação

Introdução

Principais Usos

O que não
fazer

Conclusão e Dicas

László Zsolt Garamszegi
Editor

Modern Phylogenetic Comparative Methods and Their Application in Evolutionary Biology

Concepts and Practice

 Springer

2015



Sevilha, Espanha, Novembro 2014

<http://www.mpcm-evolution.org/practice>

Apresentação

Introdução

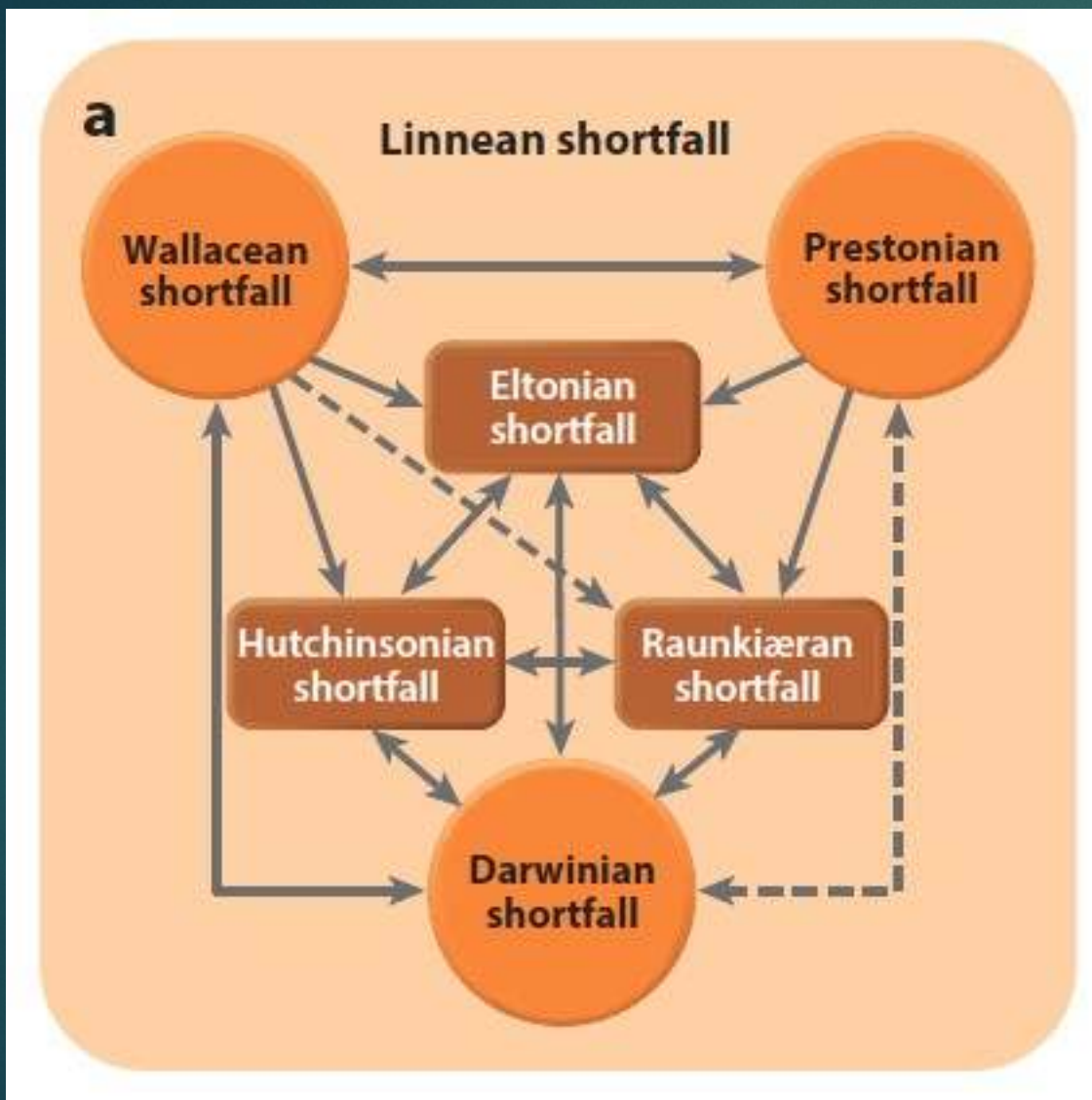
Principais Usos

O que não
fazer

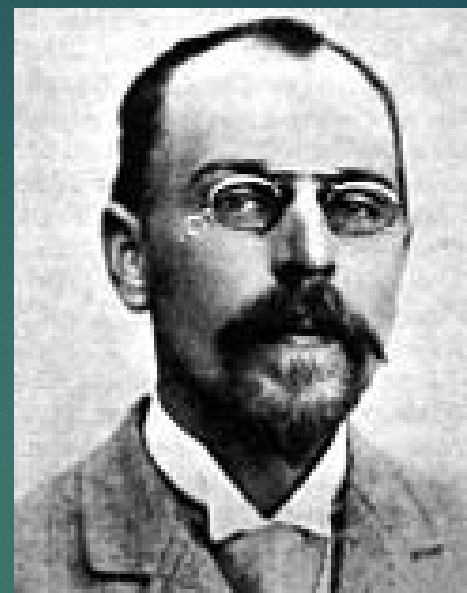
Conclusão e Dicas

Problemas mais comuns

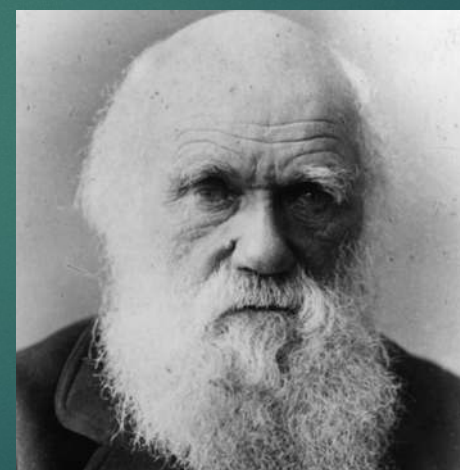
- ▶ Falta completa de dados (filogenéticos e traits)
- ▶ **Descasamento (*mismatch*) entre disponibilidade de dados de traits e de filogenias**
- ▶ Dados existem mas não estão acessíveis
- ▶ Déficits de conhecimento sobre biodiversidade: Darwiniano e Raunkiaeriano
 - ▶ Dificultam ganho de conhecimento sobre evolução de atributos
 - ▶ Impedimento para estudos comparativos



Hortal et al. 2015



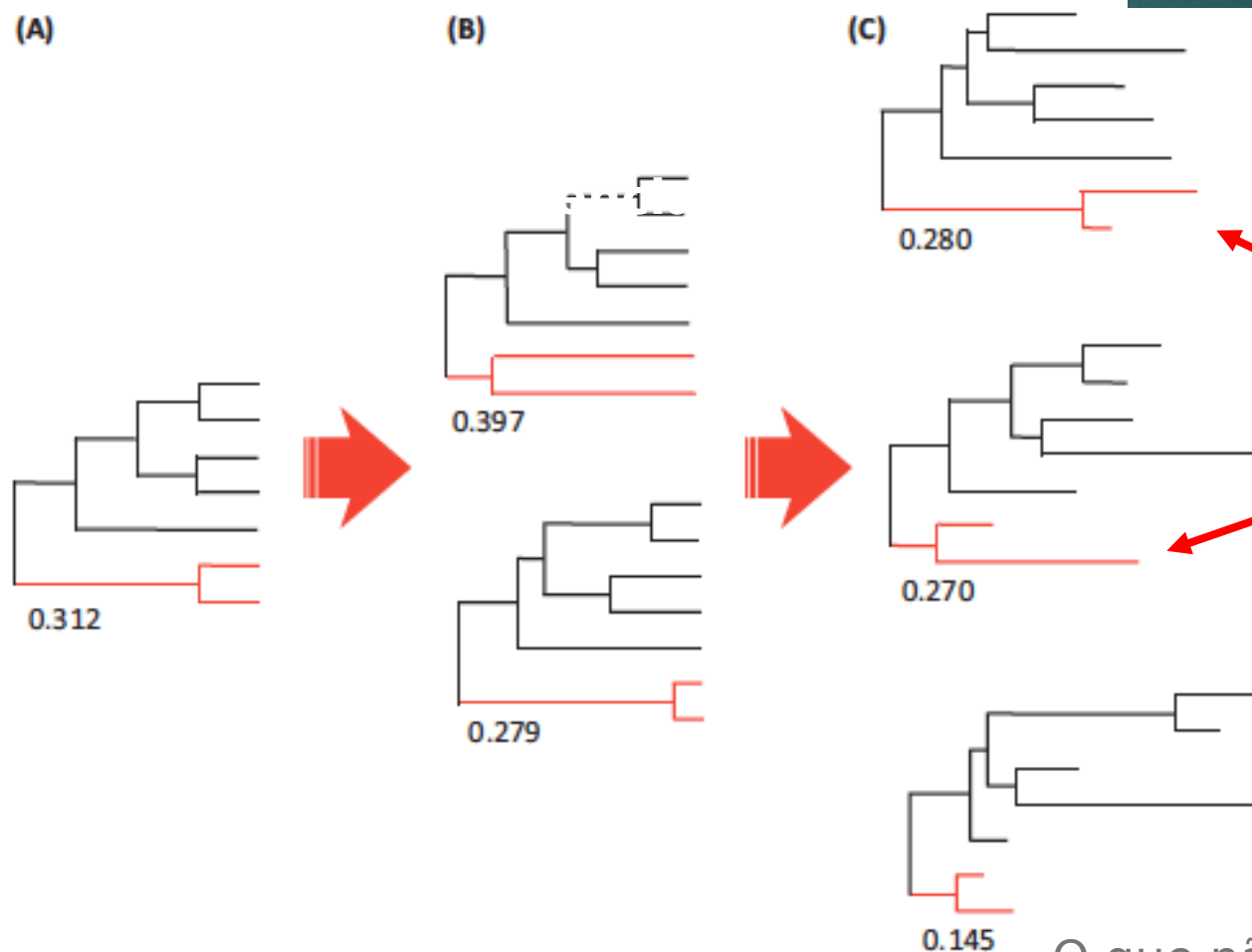
Déficit Rauniaeriano
Falta de informações
sobre atributos das
espécies



Déficit Darwiniano
Falta de informações
sobre relações entre
espécies

Darwinian shortfalls in biodiversity conservation

José Alexandre F. Diniz-Filho¹, Rafael D. Loyola¹, Pasquale Raia², Arne O. Mooers³, and Luis M. Bini¹



Modelo
evolutivo
De atributos

Ignorânci
as

MIGA SUA LOCA



SIACALME

Meu deu
agora? Me
cucua
doutorad
me soco

Pacotes phyndr e taxonlookup

Methods in Ecology and Evolution



British Ecological Society

Methods in Ecology and Evolution 2015

doi: 10.1111/2041-210X.12517

APPLICATION

A simple approach for maximizing the overlap of phylogenetic and comparative data

Matthew W. Pennell^{1,2†}, Richard G. FitzJohn^{3†} and William K. Cornwell^{4,5†}

- Maximiza a amostragem trocando espécies na árvore, para os quais não há dados de atributos, por espécies "filogeneticamente equivalentes" sem alterar a estrutura da filogenia
 - Baseado em ranking taxonômico (ou só os nomes dos gêneros) ou uma árvore topológica
 - Alternativa a métodos de aleatorização ou de imputação de dados faltantes

Ecology and Evolution

Open Access

Filling the gap in functional trait databases: use of ecological hypotheses to replace missing data

Simon Taugourdeau^{1,2,3}, Jean Villerd^{1,2}, Sylvain Plantureux^{1,2}, Olivier Huguenin-Elie³ & Bernard Amiaud^{4,5}



EDITOR'S
CHOICE

Ecography 37: 105–110, 2014

doi: 10.1111/j.1600-0587.2013.00528.x

© 2013 The Author. Ecography © 2013 Nordic Society Oikos

Subject Editor: Nathan J. Sanders. Accepted 26 July 2013

Phylogenetic imputation of plant functional trait databases

Nathan G. Swenson

Phylogenetic eigenvector maps: a framework to model and predict species traits

Guillaume Guénard^{1*}, Pierre Legendre¹ and Pedro Peres-Neto²



Genetics and Molecular Biology, 38, 3, 396–400 (2015)

Copyright © 2015, Sociedade Brasileira de Genética. Printed in Brazil

DOI: <http://dx.doi.org/10.1590/S1415-475738320140391>

Short Communication

The best of both worlds: Phylogenetic eigenvector regression and mapping

José Alexandre Felizola Diniz Filho, Fabricio Villalobos and Luis Mauricio Bini

Methods in Ecology and Evolution



Methods in Ecology and Evolution 2016

doi: 10.1111/2041-210X.12612

APPLICATION

Rphylopars: fast multivariate phylogenetic comparative methods for missing data and within-species variation

Eric W. Goolsby^{1,2,*}, Jorn Bruggeman³ and Cécile Ané^{4,5}

Methods in Ecology and Evolution



Methods in Ecology and Evolution 2014

doi: 10.1111/2041-210X.12232

Imputation of missing data in life-history trait datasets: which approach performs the best?

Caterina Penone^{1*}, Ana D. Davidson^{2,3}, Kevin T. Shoemaker², Moreno Di Marco⁴, Carlo Rondinini⁴, Thomas M. Brooks⁵, Bruce E. Young³, Catherine H. Graham² and Gabriel C. Costa¹

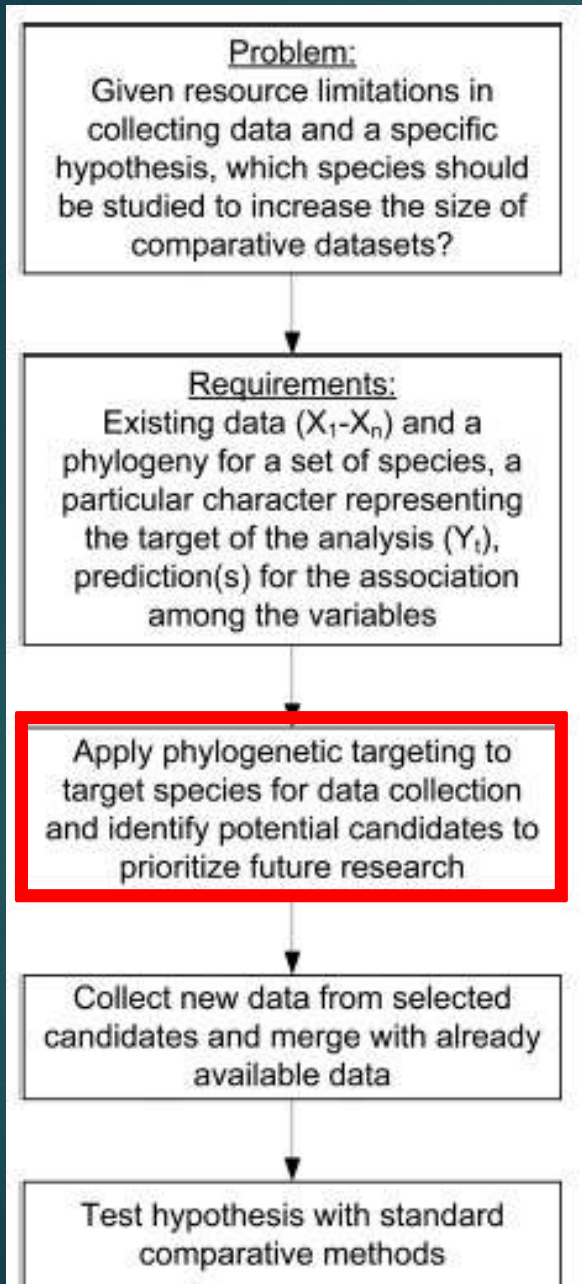
VOL. 176, NO. 5 THE AMERICAN NATURALIST NOVEMBER 2010



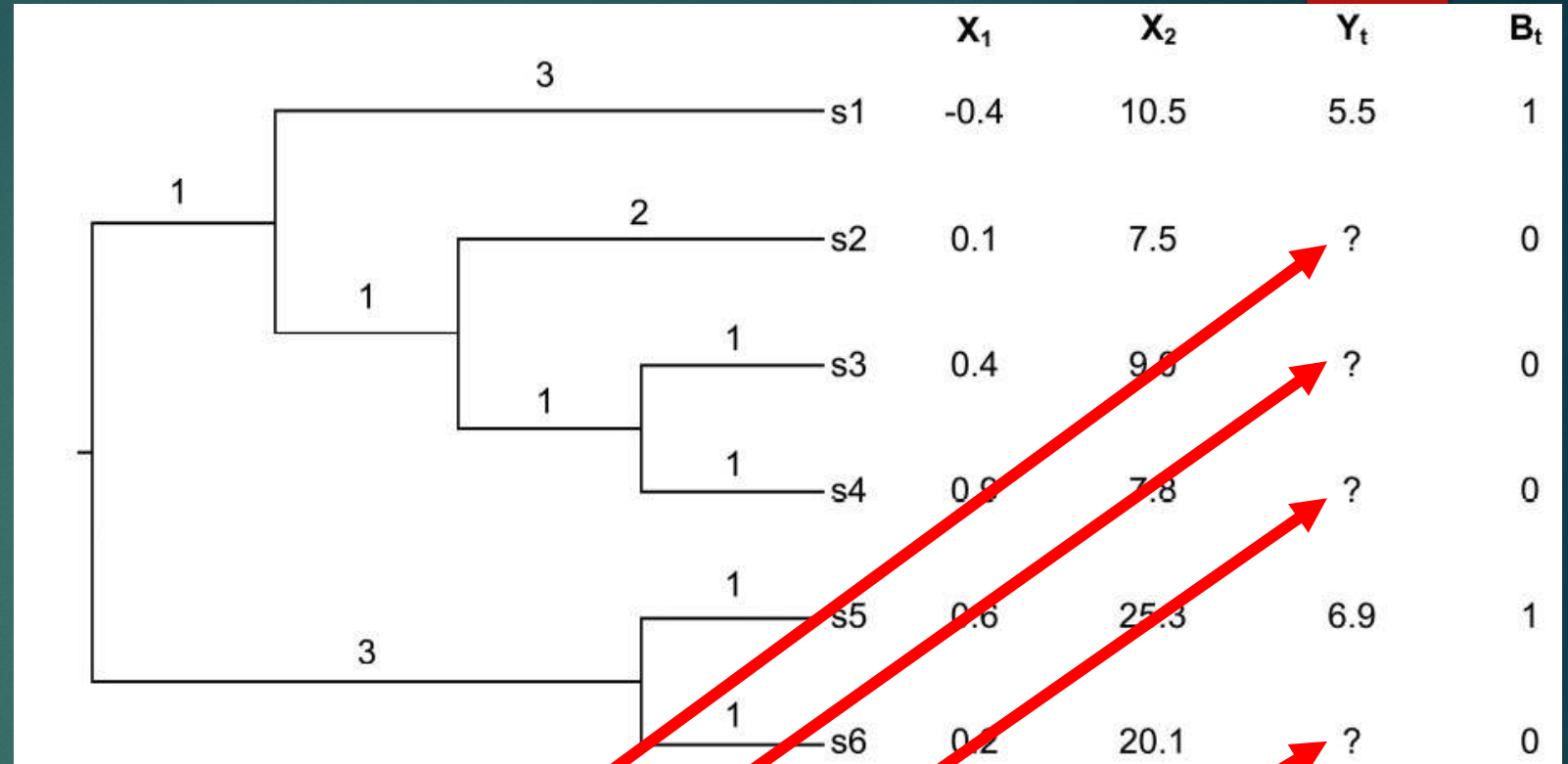
Phylogenetic Targeting of Research Effort in Evolutionary Biology

Christian Arnold^{1,2,*} and Charles L. Nunn¹

Workflow PhyloTargeting



3 Atributos contínuos



Espécies com dados
faltantes
Para variável resposta

Como otimizar o
levantamento de dados?

Limitações de PCMs

VOL. 177, NO. 6 THE AMERICAN NATURALIST JUNE 2011

Seeing the Forest for the Trees: The Limitations of Phylogenies in Comparative Biology

(American Society of Naturalists Address)*

Jonathan B. Losos[†]

Limitações de PCMs

- ▶ Muitos atributos não exibem sinal filogenético, ou seja, tem pouca correlação com a filogenia
 - ▶ Depende da escala filogenética, estatística usada etc
 - ▶ Convergência (mesmo regime adaptativo)
- ▶ Estimativas de estados ancestrais tem muita incerteza
 - ▶ Necessidade de incorporar informações de fósseis
 - ▶ Atributos evoluem sob taxas e modelos diferentes
- ▶ Filogenias são pouco úteis quando já muita variação intraespecífica
- ▶ Filogenias ajudam a desvendar padrões, e não processos
 - ▶ Velha história: inferir processos a partir de padrão
 - ▶ Mesmo padrão pode ser causado por diversos processos
 - ▶ Evite perguntas dicotômicas: "efeito relativo da filogenia (ou pior "história) e ecologia numa característica"
 - ▶ Filogenias não "causam" nada, mas sim são maneiras de representar a história evolutiva de um grupo



“Phylogenetics is a powerful tool for understanding evolution, but it is not omnipotent.”

Desusos de Métodos Comparativos

Chapter 14

Use and Misuse of Comparative Methods in the Study of Adaptation

Thomas F. Hansen

László Zolt Garamszegi
Editor

Modern Phylogenetic Comparative Methods and Their Application in Evolutionary Biology

Concepts and Practice

 Springer



Proc. R. Soc. Lond. B **205**, 581–598 (1979)

581

Printed in Great Britain

The spandrels of San Marco and the Panglossian paradigm: a critique of the adaptationist programme

BY S. J. GOULD AND R. C. LEWONTIN



Interpretação de evidências de adaptação

- ▶ Um padrão pode ser gerado por vários processos
- ▶ Só uma boa conexão entre a pergunta e o processo de interesse junto com avaliação de hipóteses alternativas permite inferir adaptação
 - ▶ Sempre levar em conta relações alométricas (com tamanho do corpo)
- ▶ Importância de se usar modelos evolutivos adequados (e.g., OU)
- ▶ Toda adaptação é fruto de seleção, mas nem toda leva à adaptação.
 - ▶ Correlação entre atributos, um estado de um atributo pode surgir como correlação a um outro e não como resposta a uma pressão evolutiva
 - ▶ Levar em conta o contexto ambiental (ou social se for o caso)



doi:10.1111/j.1420-9101.2009.01757.x

REVIEW

The seven deadly sins of comparative analysis

R. P. FRECKLETON

© 2009 THE AUTHOR. *J. EVOL. BIOL.* **22** (2009) 1367–1375

JOURNAL COMPILATION © 2009 EUROPEAN SOCIETY FOR EVOLUTIONARY BIOLOGY

Table 1 Seven deadly sins of comparative analysis.

	Problem		Consequence	Solution	Software
1	Putting undue faith in models with low R^2	Models with low explanatory power may be statistically significant. This is often a consequence of large sample sizes and in practice the effects of variables included in models are weak	The importance of weak predictors may be over-emphasized; R^2 is not a reliable measure of fit or relative fit	Use effect sizes as well as significance tests. AIC is a better measure for comparing model fit. Low R^2 is a diagnostic of lack of model fit	MasterBayes; gee function in APE; PGLM function in CAICR

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2	Reporting both PI and PC analysis	PI and PC make very different assumptions about the distribution of data, and are best regarded as alternative models for the same data. As such they should not be treated equally	Models with alternative assumptions are treated equally; potentially conflicting results may be reported	Check residuals and data for phylogenetic dependence; use a correction if appropriate	BayesTraits; GEIGER; PDAP; CAICR

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3	Not testing distributional assumptions	Phylogenetically corrected models make assumptions about the distribution of residuals that are the same as those made in nonphylogenetic analysis and are well known	Parameter estimates may be incorrect or biased. Reported P -values may be incorrect	Use conventional regression diagnostics – check for linearity, normality of residuals and homogeneity of variance (all adjusted for phylogeny)	gee function in APE; CAIC/CAICR; MasterBayes

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4	Data dredging	In analyses, comparing a large number of predictors, best fit models are selected by comparing a large number of alternative models, or by using significance tests on parameters to distinguish models	High probability of type I errors; degenerate sampling distributions for parameters. Selected model is often no better than many possible alternatives. Outcome is highly sensitive to collinearity	Clearly identify hypotheses to be tested and test those. Report all stages in the model selection process. Use the full model when appropriate; when selection is necessary use model averaging or a multi-model approach	gee function in APE; CAICR; MasterBayes

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5	Treating residuals as data	Residuals from regressions of the response on confounding variables are used to control for unwanted effects in multi-variable regressions	Results in biases, particularly when the predictors are collinear	Use multipredictor analyses rather than univariate methods; do not use residuals in model fitting	gee function in APE; CAICR; MasterBayes

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6	Ignoring alternative models	Methods such as contrasts and GLS assume that residuals are distributed according to the predictions of a Brownian model of trait evolution. This may not be the cases and other processes may be operating	The phylogenetic correction may not be fully effective. The effects of important processes such as stabilizing selection, varying rates of evolution or other factors shaping trait variation may be missed	Consider alternative models, such as OU model, δ , λ or κ transformations of Pagel (1997, 1999), or models incorporating rate variations	BayesTraits, GEIGER, LASER, APE, OUCH

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2	Reporting both PI and PC analysis	PI and PC make very different assumptions about the distribution of data, and are best regarded as alternative models for the same data. As such they should not be treated equally	Models with alternative assumptions are treated equally; potentially conflicting results may be reported	Check residuals and data for phylogenetic dependence; use a correction if appropriate	BayesTraits; GEIGER; PDAP; CAICR
3	Not testing distributional assumptions	Phylogenetically corrected models make assumptions about the distribution of residuals that are the same as those made in nonphylogenetic analysis and are well known	Parameter estimates may be incorrect or biased. Reported P -values may be incorrect	Use conventional regression diagnostics – check for linearity, normality of residuals and homogeneity of variance (all adjusted for phylogeny)	gee function in APE; CAIC/CAICR; MasterBayes
4	Data dredging	In analyses, comparing a large number of predictors, best fit models are selected by comparing a large number of alternative models, or by using significance tests on parameters to distinguish models	High probability of type I errors; degenerate sampling distributions for parameters. Selected model is often no better than many possible alternatives. Outcome is highly sensitive to collinearity	Clearly identify hypotheses to be tested and test those. Report all stages in the model selection process. Use the full model when appropriate; when selection is necessary use model averaging or a multi-model approach	gee function in APE; CAICR; MasterBayes
5	Treating residuals as data	Residuals from regressions of the response on confounding variables are used to control for unwanted effects in multi-variable regressions	Results in biases, particularly when the predictors are collinear	Use multipredictor analyses rather than univariate methods; do not use residuals in model fitting	gee function in APE; CAICR; MasterBayes
6	Ignoring alternative models	Methods such as contrasts and GLS assume that residuals are distributed according to the predictions of a Brownian model of trait evolution. This may not be the cases and other processes may be operating	The phylogenetic correction may not be fully effective. The effects of important processes such as stabilizing selection, varying rates of evolution or other factors shaping trait variation may be missed	Consider alternative models, such as OU model, δ , λ or κ transformations of Pagel (1997, 1999), or models incorporating rate variations	BayesTraits, GEIGER, LASER, APE, OUCH
7	Ignoring quality control of data	Data from disparate sources vary in quality and may be erroneous. Data may be missing for significant numbers of species	Low quality data will compromise statistical power. Missing data can lead to biases in the outcome of analyses	Employ quality criteria for data inclusion. Analyse data to determine whether data are missing randomly with respect to other variables. Consider imputation methods	MasterBayes

Colocar muita ênfase em modelos com R^2 baixo

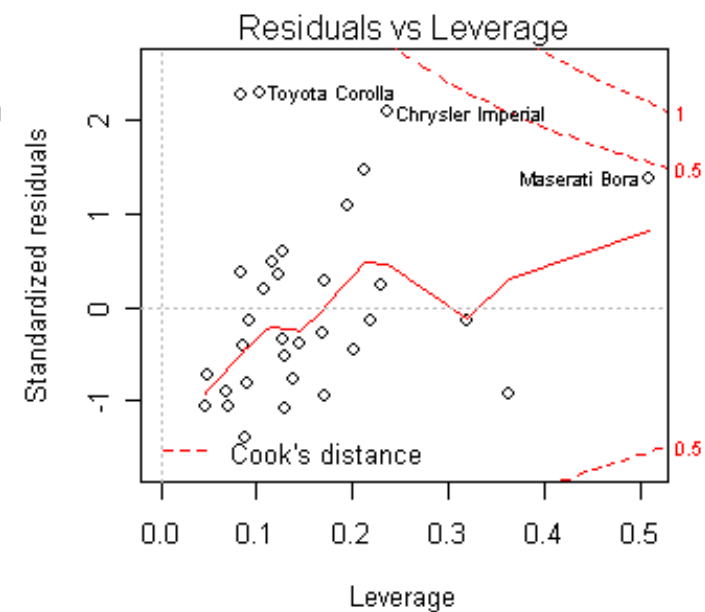
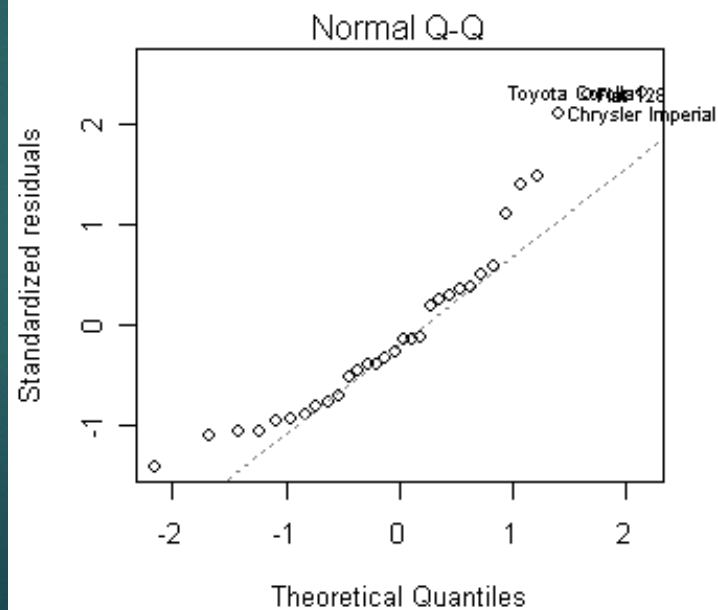
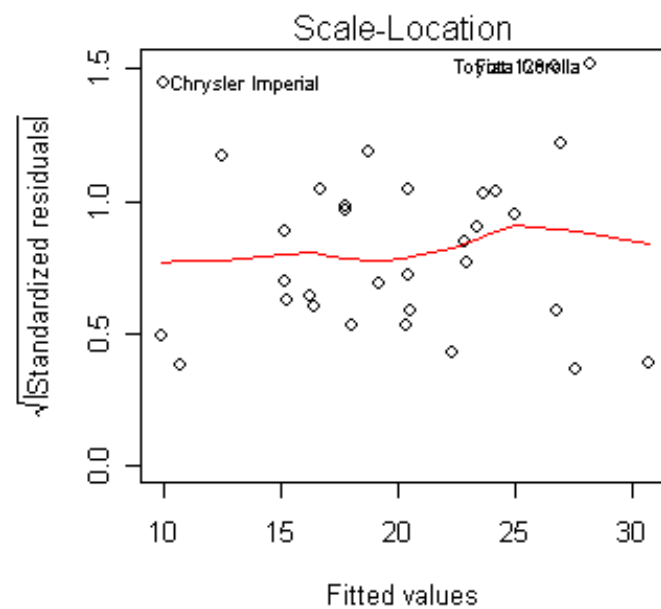
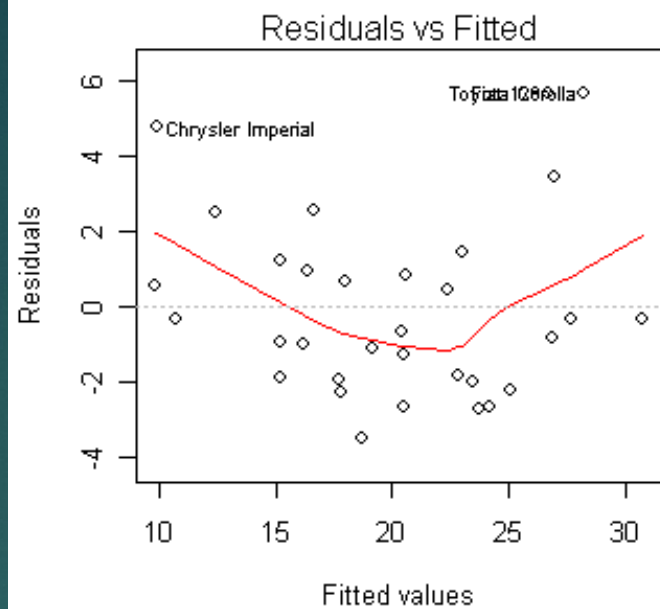
- ▶ R^2 = proporção de variância explicada/variância nos dados
- ▶ Não pesa pela quantidade de variáveis
- ▶ Reportar tamanho do efeito e forma da relação entre variáveis
- ▶ Utilizar estatísticas de Teoria da Informação (e.g., AIC, wAIC, BIC, Deviance etc) que pesam o número de variáveis e informam o peso de evidência de um modelo

Reportar resultados de análises filogenéticas e tradicionais

- ▶ Se o objetivo da análise é comparar dados de diferentes espécies e detectamos que há dependência entre elas, obrigatoriamente deve-se usar métodos que levam essa dependência em conta
- ▶ Análise filogenéticas e comuns não estimam os mesmos parâmetros e por isso não são comparáveis
- ▶ Especialmente problemático em modelos lineares, tipo (P)GLS
 - ▶ Solução: estimar um parâmetro (λ de Pagel) que controle a autocorrelação filogenéticos nos resíduos

Não testar pressupostos

- ▶ Diagnosticar o modelo para detectar violações dos pressupostos
- ▶ Autocorrelação, normalidade e homogeneidade de variância **dos erros**



Ajustando um PGLS

- ▶ O que é melhor: ajustar um modelo em que simultaneamente se estima o "sinal" filogenético dos resíduos (usando ML) e se estima os parâmetros do modelo PGLS, p.ex. usando o lambda do Pagel

Methods in Ecology and Evolution



British Ecological Society

Methods in Ecology & Evolution 2010, 1, 319–329

doi: 10.1111/j.2041-210X.2010.00044.x

Phylogenetic signal and linear regression on species data

Liam J. Revell*

Data dredging

- ▶ Confundir Análise de dados Exploratória com Análise inferencial
 - ▶ Errado: Apresentar uma lista de modelos com todas as combinação possíveis de variáveis
 - ▶ Correto: definir *a priori* modelos biologicamente plausíveis e testar somente estes

Tratar resíduos como dados primários

- ▶ Errado: Utilizar resíduos de, por exemplo, uma regressão entre tamanho de corpo e frequência da vocalização para trabalhar em outra análise
 - ▶ Correto: utilizar a variável "confundidora" como co-variável numa análise (e.g., ANCOVA)

Ignorar modelos alternativos de evolução

- ▶ Não testar qual modelo de evolução melhor explica seus dados
- ▶ Usar parâmetros "default" do programa ou que assumem Movimento Browniano
- ▶ Solução: ajustar modelos de evolução alternativos (e.g., OU, EB/LB etc)

Não avaliar qualidade dos dados

- ▶ Se dados vêm de várias fontes (e.g., bases de dados da internet), é importante avaliar a qualidade (reliability) dos mesmos antes de realizar análises
- ▶ Decidir de usar ou não métodos para imputar dados faltantes ou se apaga o dados completamente
- ▶ O mesmo vale para a filogenia
 - ▶ Filogenias são hipóteses que podem ser muito ou pouco sustentadas, é importante incorporar, sempre que possível, incerteza filogenética nas análises

Shedding light on the ‘dark side’ of phylogenetic comparative methods

Natalie Cooper^{1,2*}, Gavin H. Thomas³ and Richard G. FitzJohn⁴

- ▶ Leitura cuidadosa antes de utilizar os métodos no R
- ▶ Sempre ser crítico sobre os resultados
- ▶ Importância de se reportar as análises utilizando ferramentas de reprodutibilidade (R Markdown, LaTeX etc)

Onde encontrar mais informações sobre PCMs?

- ▶ Canais do YouTube
 - ▶ Evolution 2015, 2016 (Vídeos com palestras do congresso)
 - ▶ PhyloMeth (Curso online sobre métodos comparativos)
 - ▶ NMBios (cursos sobre genética quantitativa e métodos comparativos)
 - ▶ Phyloseminar (ciclo de palestras online sobre diversos temas em ecologia e evolução)
- ▶ Listas de email (R-SIG-PHYLO) e foruns (stackoverflow.com)
- ▶ Cursos do Liam Revell e Luke Harmon (disponíveis no GitHub)
- ▶ Companion sites de livros do Charles Nunn e Garamszegi



“ Obrigado pela atenção!

Perguntas?

”

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